1.6 Bonding

Students will be assessed on their ability to:

1 Ionic bonding

a recall and interpret evidence for the existence of ions, limited to physical properties of ionic compounds, electron density maps and the migration of ions, e.g. electrolysis of aqueous copper chromate(VI)

b describe the formation of ions in terms of electron loss or gain

c draw electron configuration diagrams of cations and anions using dots or crosses to represent electrons

d describe ionic crystals as giant lattices of ions

e describe ionic bonding as the result of strong net electrostatic attraction between ions

f recall trends in ionic radii down a group and for a set of isoelectronic ions, e.g. N^{3-} to Al^{3+}

g recall the stages involved in the formation of a solid ionic crystal from its elements and that this leads to a measure value for the lattice energy (students will not be expected to draw the full Born-Haber cycles)

h test the ionic model for ionic bonding of a particular compound by comparison of lattice energies obtained from the experimental values used in Born-Haber cycles, with provided values calculated from electrostatic theory

i explain the meaning of the term *polarisation* as applied to ions

j demonstrate an understanding that the polarising power of a cation depends on its radius and charge, and the polarisability of an anion depends on its size

k demonstrate an understanding that polarisation of anions by cations leads to some covalency in an ionic bond, based on evidence from the Born-Haber cycle

l use values calculated for standard heats of formation based on Born-Haber cycles to explain why particular ionic compounds exist, e.g. the relative stability of $MgCl_2$ over $MgCl$ or $MgCl_3$ and $NaCl$ over $NaCl_2$.

2 Covalent bonding

a demonstrate an understanding that covalent bonding is strong and arises from the electrostatic attraction between the nucleus and the electrons which are between the nuclei, based on the evidence:

i the physical properties of giant atomic structures

ii electron density maps for simple molecules

b draw electron configuration diagrams for simple covalently bonded molecules, including those with multiple bonds and dative covalent bonds, using dots or crosses to represent electrons.

3 Metallic bonding

a demonstrate an understanding that metals consist of giant lattices of metal ions in a sea of delocalised electrons

b describe metallic bonding as the strong attraction between metal ions and the sea of delocalised electrons

c use the models in 1.6.3a and 1.6.3b to interpret simple properties of metals, e.g. conductivity and melting temperatures.